MAHADEV GOLD

14/10/2026

CO-1 AT-3

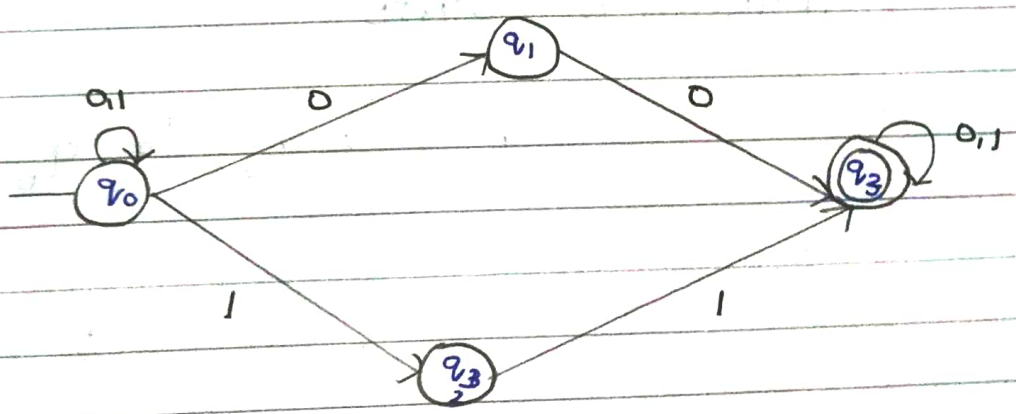
1. Design a nondeterministic Finite automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are same (either 00 or 11) Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA

AIM: To design an NFA over the alphabet $\Sigma = \{0,1\}$ that accepts all binary strings whose last two strings are identical.

NFA with 3 states:

q_0 : Initial state
q_1 : If a is 0
q_2 : If a is 1

STATE DIAGRAM



An NFA is defined by 5 tuple.

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{ q_0, q_1, q_2, q_f \}$

$\Sigma = \{ 0, 1 \}$

δ = TRANSITION FUNCTION:

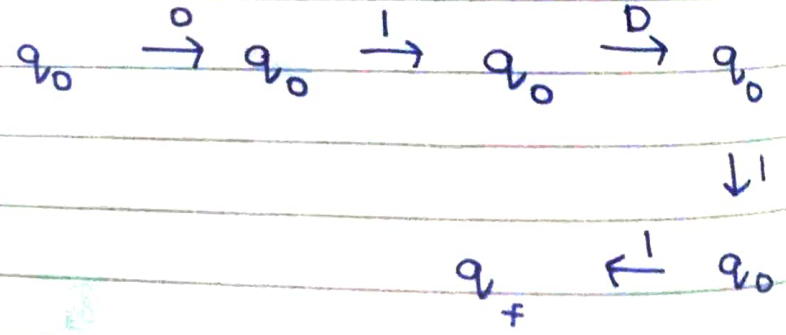
STATE	INPUT 0	INPUT 1
q_0	$\{ q_0, q_1 \}$	$\{ q_0, q_2 \}$
q_1	$\{ q_f \}$	ϕ
q_2	ϕ	$\{ q_1 \}$
q_f	ϕ	ϕ

(10) q_0 = INITIAL STATE

$F = \{ q_f \}$

Example input string: $w = 01011$

TRANSITION STATE:

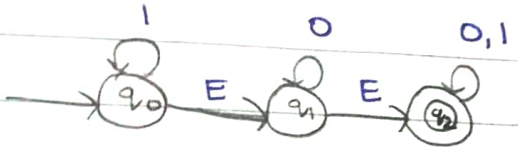


Since the string ends in the final state q_f, the string 01011 is accepted by the NFA

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If the input string is 1010, 1101 the string gets rejected.

5. A NFA with ε - transitions is given below. Find ε closure for each state and design an equivalent DFA which accepts the same language. Draw it's transition table and mark the initial and final states.



From the image, ε - NFA has 3 states : q_0, q_1, q_2

(10)

Initial state : q_0
Final state : q_2

The ε - closure includes the states itself and every state reachable through ε - transitions without consuming any input.

* ε - closure $(q_0) = \{q_0, q_1, q_2\}$
* ε - closure $(q_1) = \{q_1, q_2\}$
* ε - closure $(q_2) = \{q_2\}$

DFA states are:

$A = \{q_0, q_1, q_2\}$ - initial states

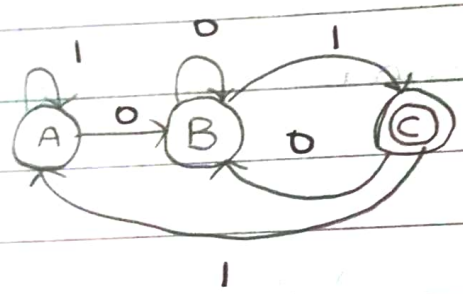
$B = \{q_1, q_2\}$

$C = \{q_2\}$

DFA transition table:

DFA state	Input 0	Input 1
* A	B	A
* B	B	C
* C	C	C

DFA transition diagram:



Thus, the equivalent DFA is constructed using the ε-closure method. Since all reachable DFA are final states, the automata accepts every binary string, including empty string ε